Howard Huang

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Mechanical Engineering student specializing in CAD, simulation work, and programing.

Education:

Bachelor of Science, Mechanical Engineering **May 2028**
Georgia Institute of Technology, Atlanta, GA **--/4.0**
Rose-Hulman Institute of Technology, Terre Haute, IN **3.8/4.0**
Related Courses: Mechatronics, Design for Manufacturing, Materials Engineering, Thermodynamics, Numerical Methods, Advanced Design of Mechanisms, Machine Component Design, System Dynamics

Skills:

Design and Simulation: Ansys, SolidWorks, Autodesk Inventor, Arduino/SFRs, XFLR5, PicoSoft, SEM
Manufacturing: CNC, lathe, plasma cutting, sandcasting, welding, additive manufacturing (FDM + nanoscale 2PP), safety
Software: Python, Java, Arduino IDE, Simulink, GitHub, Word, Excel, PowerPoint, SAP 2000, MATLAB + Animation

Research Experience:

Rose-Hulman Research – *Student Researcher* **Aug 2025 - Sep 2026**
- Investigate how lateral center-of-mass shifts **(±5 cm)** affect **trim authority** in coordinated banked turns **(0.6 m UAV)**
- Built a **Python 6-DOF nonlinear trim solver** using interpolated AVL aerodynamics maps
- Generated **101,871-case AVL aerodynamic database** across angle of attack, aileron, elevator, and rudder deflections
- Validated solver predictions against AVL points and quantified COM shift versus required aileron deflection
- Analysis of the control input difference of COM shifted mass acting as a **input control mechanism**
- Submitted Technical abstract To **AIAA SciTech Conference** and preparing for conference under mentorship of Dr. Kevin P.T Haughn

Caltech STEM Research – *Student Researcher* **Jul 2023 - Jun 2024**
- Fabricated **~500 nm architected materials** via **nTopology** + two-photon polymerization using nanoscribe
- Analyzed **15+ microstructures (SEM),** identifying failure mechanisms and individual material properties
- Quantified **2–4× stiffness variation** (yield strength ~0.5-2 GPa) driven purely by geometry
- Synthesized experimental + theoretical results into a **technical poster**, communicating advanced materials behavior
- Led by Wenxin Zhang under the lab of Dr. Julia R, Greer

Project Experience:

FIRST Robotics Competition – *Engineering VP* **Aug 2020 - Jun 2024**
- Directed development of **3 competition robots (**500+ components each) under aggressive 6-week design cycles
- Led **10–25 member interdisciplinary team**, coordinating mechanical, electrical, and software subsystems
- Drove **4+ iterative design cycles** per subsystem, optimizing performance, manufacturability, and reliability
- Reduced weight by 8% and cut costs by $2,000 through material optimization and improved fabrication processes
- Achieved first-ever **Regional Championship + World Championship** qualification scaling team by +20 members

Two-Player Pinball Game (Mechatronic Course) **Sep 2025 - Dec 2025**
- Engineered **50+ component integrated system** combining mechanical actuation, sensing, and embedded control
- Developed **Arduino-based** real-time control system for responsive gameplay
- Implemented finite state machine **(FSM)** governing scoring logic, timing, and multiplayer interaction
- Executed **3 full design iterations**, refining system reliability and user experience
- Validated performance across **100+ test cycles**, ensuring repeatability and robustness

Startup Weekend (Techstars) – *Entrepreneurship Startup* **Oct 2024 - Dec 2024**
- Developed scalable **business model** for an EEG-based diagnostic device for depression patients
- Modeled cost structure, revenue pathways, and customer segmentation
- Conducted **stakeholder interviews** with industry professionals, informing product-market fit and deployment strategy

Additional Information:

Additional Projects: SolidWorks Humanoid CAD (200+ parts, kinematics), Rose Grand Prix (SAE), Rose Rocketry, Java Game (multi-level, collision logic), $4 Guitar (ultra-low-cost design, sustainability), Metallography (4 bolt samples, microstructure analysis
Hobbies: Aviation, Traveling, Stock Trading, Gym, Guitar, Water polo,